Geonwoo Kim

UNDERGRADUATE | COMPETITIVE PROGRAMMER | DEVELOPER | RESEARCHER

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EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST)
School of Undergraduate Studies
GPA: 4.21 / 4.30
Daegu, South Korea
Feb 2026 - Present

Daejeon Science High School for the Gifted
High School Diploma
GPA: 3.70 / 4.30
Daejeon, South Korea
Mar 2024 - Jan 2026

COMPETITIVE PROGRAMMING

UCPC 2026 Final Round
Collegiate Team Programming Contest
Finalist - 28th of 60 teams; 149 preliminary teams
2026

Korea Olympiad in Informatics (KOI)
Algorithms and Data Structures
National Competition
2017 - 2025

- National Silver Prize in four consecutive years (2017-2020) and Regional Grand Prize in 2017 and 2018.
- In the High School Division: Silver and Bronze in the 2024 rounds; Bronze and Honorable Mention in both 2023 and 2025.
- Completed the 33rd KOI Summer School (2024) and the 34th KOI Winter School (2025).

Nexon Youth Programming Challenge (NYPC)
National Programming Competition
Finalist, Ages 15-19
2024

RESEARCH

DGIST CTA Lab
Computational Theory and Applications Lab
Undergraduate Researcher
Sep 2026 - Present

- Participating in research on algorithm theory.

Self-Correction Quantization for Korean Large Language Models
Daejeon Science High School Graduation Research Program
Proposed Self-Correction Quantization (SCQ)
Sole Researcher
Mar 2025 - Nov 2025

- Proposed Self-Correction Quantization (SCQ), which fine-tunes a quantized model on self-generated corrections to recover Korean linguistic capability without an external correction corpus.
- Compared PTQ, generic QAT, and SCQ on Korean LLMs up to 12.8B parameters: 4-bit PTQ reduced memory by about 84%, while SCQ improved the grammar-sensitivity score by 2.75x over PTQ with less general-reasoning degradation than generic QAT.

3D Gaussian Splatting for Metaverse Content Development
Science Gifted Creative Research (R&E)
Grand Prize and Minister of Science and ICT Award
Team Lead
Mar 2024 - Dec 2024

- Designed 33 controlled dataset conditions varying image count, brightness, and contrast; built a Python pipeline that automated training-data generation and evaluation with PSNR, SSIM, and LPIPS.
- Used frequency-energy and local-contrast analyses to explain reconstruction-quality degradation and identify robust capture conditions.

EXPERIENCE

KimuStory - Ep
Main Developer
Feb 2022 - Present

Production Discord Fishing Game
530,000+ users across 53,000+ Discord servers

- Build and maintain a stateful game service used by 530,000+ unique users across 53,000+ Discord servers.
- Engineer interaction, progression, and backend systems with TypeScript, Node.js, discord.js, Redis, NestJS, REST APIs, and PostgreSQL, emphasizing concurrency, correctness, and fault handling.

PROJECTS

Sangsaeng-ieum - AI Support Platform for Foreign Workers
Team Lead
Mar 2025 - Jun 2025

AI Convergence Policy Discovery Hackathon
Honorable Mention

- Led a four-person team in prototyping AI-assisted worker matching, a multilingual support chatbot, and guided electronic contract and administrative workflows.

SnowMix - AI-Assisted Web Accessibility
Team Lead and Lead Developer
Jul 2024 - Nov 2024

Web Accessibility Project
National Bronze Prize

- Built a browser extension and AI service that analyzes visual structure and dynamic content, produces navigable descriptions, and transforms images for color-blind users.
- Led a four-person team across planning and implementation using JavaScript, Python, and PyTorch.

ARKit Indoor Navigation for Visually Impaired Users
Developer
Jun 2024 - Sep 2024

iOS Prototype for Spatial Mapping and Audio Route Guidance
Accessibility-Aware Indoor Navigation

- Constructed a 3D graph from ARKit spatial anchors and implemented A* pathfinding with accessibility-aware edge weights for step-by-step audio navigation.
- Mapped and repeatedly tested dozens of points in a school building, refining coordinate handling and calibration to reduce sensor-induced routing errors.

LEADERSHIP AND ACTIVITIES

Pineapple Computer Club
President (2024)
Apr 2023 - Aug 2025

Information Technology Club
Revitalized a long-running school club

- Revitalized a long-running information technology club by organizing regular activities across programming, software projects, and broader computing topics.

Infomaker
Founder and President
Mar 2024 - Dec 2025

Algorithm Problem-Solving Club
Led about 20 students

- Founded the school's first algorithm problem-solving club and led about 20 students through lessons, contest practice, and peer solution reviews.
- Built a solved.ac-integrated practice and ranking system that generated workbooks by topic and adapted problem selection to different skill levels.

DSHStack Programming Contests
Organizer, Problem Setter, and Reviewer
2024 - 2025

Student-Led Algorithm Competitions
Up to 180 external participants

- Organized contests for roughly 40-60 in-school participants and about 180 external participants per event.
- Coordinated problem setting, generators, validators, adversarial tests, seven external reviewers, editorials, and live solution sessions using Polygon and testlib.

HONORS AND AWARDS

National and Regional Awards - Korea Olympiad in Informatics (KOI) 2017 - 2025

Honorable Mention - AI Convergence Policy Discovery Hackathon Jun 2025
Sangsaeng-ieum: An AI-Based Support Platform for Foreign Workers — Provincial Governor's Award

Grand Prize - Science Gifted Creative Research (R&E) Jan 2025
Developing Metaverse Content through 3D Gaussian Splatting — Minister of Science and ICT Award

Bronze Prize - 9th National High School Club Software Competition Nov 2024
SnowMix: An AI Web Accessibility System for Visually Impaired Users — University President's Award

Bronze Prize - 30th Daejeon Student Science Exploration Olympiad Sep 2024
Exploring Applications of Stable Diffusion — Superintendent of Education Award

ADDITIONAL EDUCATION

Jane Street FTTP 2026 Hong Kong
Trading and Technology Program for First-Year Undergraduates — Sep 30 - Oct 3, 2026

- Selected for and participated in classes on trading and technology and a team-based mock trading simulation, exploring how mathematics and technology inform trading.

Information Technology Gifted Education Program Daejeon, South Korea
Gifted Education Center, Daejeon Education Information Institute — 2018 - 2020
Beginner, Intermediate, and Advanced Courses

- Built a C++ Korean typing tutor (2018), a Python/Qt vocabulary-learning application using the Open Korean Dictionary API (2019), and a KoGPT2 conversational bot (2020).